



CLIMACROP INTELLIGENCE

Project Documentation

“Transforming climate and market data into actionable agricultural and financial decisions.”

Climate-Smart Decision Support & Agri-Fintech De-Risking Platform for Kenya

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KPC INUKA Capstone Project

github.com/CarsonSila/ClimaCrop-Intelligence

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1. Executive Summary

ClimaCrop Intelligence is a climate-smart decision support and agri-fintech de-risking platform designed to address the growing challenges of climate volatility facing smallholder agriculture in Kenya. It combines localized 10-year climate patterns, a 40-crop suitability matrix, and institutional credit-underwriting logic into a single, integrated tool.

The platform serves three primary stakeholder groups — agricultural cooperatives, financial institutions (banks and SACCOs), and researchers — by translating raw climate, crop and market data into practical agricultural and financial decisions, rather than simply reporting weather conditions or prices.

What sets the platform apart is its commitment to transparency and trust. Every recommendation, risk score and financial output carries a clear provenance tag — measured, official, estimated, modeled, or assumed — so users always know exactly how much confidence to place in what they are seeing. Role-based access ensures cooperative members, bank officers, researchers and administrators each see a dashboard tailored to their needs, built on the same underlying analytical engines.

116

Weather Stations

35

Counties

40

Crops

5

Markets

10 yrs

Climate Data

2. Problem Statement & Solution Overview

2.1 The Problem We're Solving

Smallholder agriculture forms the backbone of Kenya's economy, contributing over 33% of national GDP and employing 70% of the rural population. Climate variability — shifting seasonal rain patterns, extended dry spells and rising temperatures — has made traditional farming calendars increasingly unreliable, leaving farmers guessing about what to plant, when to plant, and whether it will pay off. At the same time, fluctuating agricultural prices make it difficult to identify favourable markets, and these same uncertainties make it harder for banks and SACCOs to price agricultural lending fairly.

Traditional weather applications only answer one question: “Will it rain tomorrow?” ClimaCrop Intelligence is built to answer deeper questions that farmers, cooperatives and financial institutions actually need answered:

- Farmers & cooperatives: Given local climate trends, what should we plant this season, what yield and profit can we expect per acre, and which regional market offers the best price?
- Banks & SACCOs: What is the composite climate and agricultural risk of financing this farmer or cooperative? What loan facility are they eligible for, and what should the risk-adjusted interest rate and expected default probability be?
- Researchers: How do ten years of station-level rainfall and temperature data compare against a transparent, rule-based advisory model versus a probabilistic machine-learning model?

2.2 Our Solution

ClimaCrop Intelligence answers all of these questions through a single, integrated platform that combines:

- Climate intelligence from 116 weather stations across 35 Kenyan counties, providing 10-year seasonal aggregates for both Long Rains (MAM) and Short Rains (OND).
- Agronomic expertise encoded into a 40-crop database spanning cereals, pulses, roots & tubers, horticulture and cash crops, each with specific rainfall, temperature and economic requirements.
- Market awareness through pricing data from five major regional markets: Nairobi,

Nakuru, Eldoret, Kisumu and Mombasa.

- Financial modeling that translates climate, crop and market signals into risk scores, loan recommendations and portfolio simulations.
- KilimoBot, a conversational AI assistant that makes the platform's insights accessible to users of all technical backgrounds.

The project focuses on translating climate information into practical agricultural and financial decisions — not simply providing another weather forecast.

3. Target Users & Value Delivered

Agricultural Cooperatives

Use the platform to understand crop suitability for their members and identify the market opportunities offering the best net return.

Banks

Use agricultural and climate risk intelligence to support agricultural financing decisions with a defensible, data-backed risk score.

SACCOs

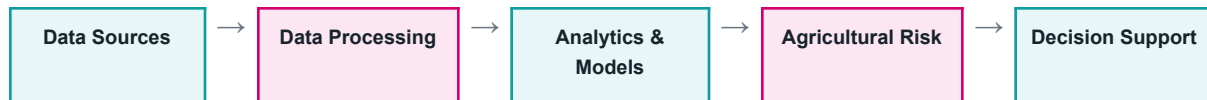
Use the same risk information to better understand farmer and agricultural loan exposure across their portfolio.

Researchers

Use the climate, crop and market analysis — including a side-by-side rule-based vs. machine-learning comparison — for agricultural and environmental research.

4. System Architecture

The system follows a layered, data-to-decision architecture that ensures separation of concerns, maintainability and extensibility. Each layer builds on the one below it, creating a clean flow from raw data to actionable insight.



4.1 The Five Architectural Layers

Layer 0 — Authentication & Authorization

- Four distinct personas: Cooperative Member, Bank/SACCO Officer, Researcher and Admin.
- SHA-256 salted password hashing for credential storage, backed by a JSON user store.
- Each role sees a tailored subset of tabs and a personalized landing dashboard.

Layer 1 — Data Ingestion

- 116 TAHMO weather stations mapped to 35 Kenyan counties.
- 10-year seasonal aggregates for rainfall, temperature and drought risk.
- A 40-crop agronomic and economic matrix informed by KNBS and FAO baselines.
- A 5-market regional pricing database covering all 40 crops.
- Every value is wrapped with a provenance tag for transparency.

Layer 2 — Advisory & Analytics Engines

- Climate pattern clustering via K-Means (k=3) identifying Normal, Flood and Dry patterns across 559 county-season records.
- Crop suitability assessment in two modes: a transparent rule-based AEZ engine (default) and a legacy Random Forest model (kept for audit and comparison).
- Market arbitrage and price forecasting with transport-adjusted net-margin ranking.

Layer 3 — Fintech & Decision Logic

- Cooperative sizing: total CapEx, net farm profit, and Benefit-Cost Ratio (BCR).

- Composite risk underwriting combining climate, crop-fitness and market volatility.
- Facility sizing and risk-adjusted interest-rate pricing.
- A portfolio simulator for stress-testing an entire loan book.

Layer 4 — User Interface

- A role-aware Streamlit application with personalized dashboards per persona.
- KilimoBot, an AI assistant powered by Google Gemini with an offline fallback.
- Interactive visualizations of climate trends, crop scores and market data.

5. Data & Data Models

The platform combines three major data categories — climate, crop and market — processed through a consistent validation and feature-engineering pipeline before reaching the analytics layer.



5.1 Climate Data Model (county_climate_historical.csv)

559 county-season records across 35 counties. Each record captures:

Field	Description	Example
county	Kenyan county name	Kiambu
season	Rainfall season (MAM or OND)	MAM
seasonal_rainfall_mm	Average seasonal rainfall (mm)	450.2
rainy_days	Number of rainy days in the season	62
heavy_rain_days	Days with heavy rainfall (>20mm)	8
max_dry_spell_days	Longest consecutive dry period	14
onset_week	Week the rainy season typically begins	12
temp_mean_c / max_c / min_c	Average / max / min temperature (°C)	22.5 / 28.1 / 16.9
elevation_m	Elevation (metres)	1,750
drought_risk_index	Composite drought risk indicator	0.35

5.2 Crop Database Model (crops_database.csv)

40 crops across five commercial classes, each with agronomic thresholds (min/max rainfall and temperature, growth days, drought tolerance), economic parameters (cost per acre, yield per acre, base price) and a provenance / confidence tag.

Category	Crops Included
Cereals (6)	Maize, Sorghum, Finger Millet, Pearl Millet, Wheat, Rice
Pulses / Legumes (6)	Common Beans, Cowpeas, Green Grams, Pigeon Peas, Soybeans, Groundnuts
Roots & Tubers (4)	Irish Potatoes, Sweet Potatoes, Cassava, Arrowroots
Horticulture (12)	Tomatoes, Onions, Cabbage, Kales, Spinach, Capsicum, French Beans, Carrots, Watermelon, Butternut Squash, Pumpkin, Garden Peas
Cash & Tree Crops (12)	Coffee, Tea, Sugarcane, Avocado, Mangoes, Bananas, Macadamia, Cotton, Pyrethrum, Cashew Nuts, Passion Fruit, Miraa

5.3 Market Price Model (market_prices.csv)

200 records (40 crops × 5 markets), capturing:

Field	Description
crop / market	Crop name and one of five regional markets
base_price	Reference base price (KES per kg)
market_price	Current market price (KES per kg)
peak_harvest_price / off_season_price	Seasonal price extremes
volatility_cv	Coefficient of variation for price volatility

5.4 Data Provenance & Confidence

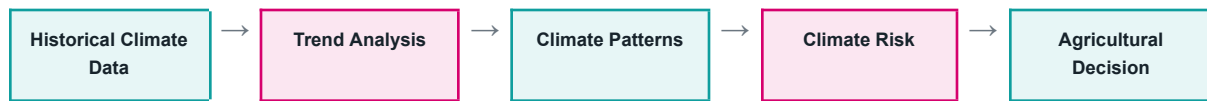
Every value the system reports is wrapped as a SourcedValue carrying a provenance tag. A ProvenanceReport aggregates these into an overall confidence score, rendered as a plain-language caveat so users never mistake modeled or assumed data for authoritative, measured data.

Tag	Meaning	Confidence Weight
MEASURED	Directly observed data (e.g. a station reading)	1.00
OFFICIAL	From an official published source	0.90
ESTIMATED	Estimated via a documented method	0.65

MODELED	Output of a model (e.g. the legacy ML classifier)	0.50
ASSUMED	Unverified placeholder or default value	0.20

6. Climate Intelligence

ClimaCrop analyses historical climate information to identify patterns that affect agricultural decisions, including temperature trends, rainfall patterns and variability, dry spells, drought risk, and regional climate differences.



Climate pattern clustering uses K-Means ($k=3$) across 559 county-season records to classify each season as Normal, Flood-risk or Dry-risk — the foundation the rest of the pipeline builds on. The climate component helps users understand how environmental conditions vary across locations and periods, before a single crop or loan decision is made.

7. Crop Suitability Intelligence

The crop suitability component evaluates environmental conditions against crop requirements to determine which of the 40 crops are best suited to a given location and season, considering rainfall, temperature, humidity, location and season.



7.1 Rule-Based Engine (Default)

The rule-based engine is the system's primary source of truth. It checks a crop's agronomic thresholds (minimum/maximum rainfall, temperature, growth days) against local climate data, producing a 0–100 suitability score as a weighted blend of rainfall suitability, temperature suitability and drought tolerance.

- Completely transparent and auditable — no training data required.
- Scores are tagged OFFICIAL or ESTIMATED provenance.
- Directly tied to established agronomic knowledge (AEZ thresholds).

7.2 Machine Learning Engine (Legacy / Audit)

A Random Forest classifier is kept for side-by-side comparison and benchmarking. It was trained on synthetic samples generated from the same agronomic thresholds the rule-based engine uses — a circularity that means it is effectively learning to mimic its own training data

rather than discovering independent patterns.

- Scores are tagged MODELED with a lower confidence weight (0.50).
- Not independent of the rule-based engine, and not treated as a more advanced alternative.
- Retained for researcher benchmarking and stakeholder trust-building, available as a sidebar toggle.

8. Market Intelligence

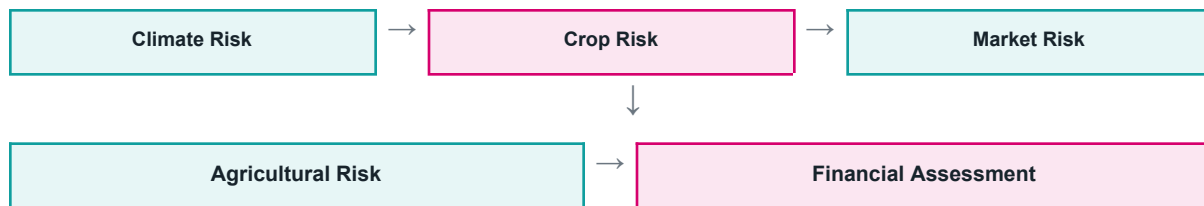
The market component analyses agricultural prices across five selected markets to surface historical price trends, market comparisons, crop-market relationships and potential market opportunities.



Net margins are transport-adjusted, so the engine ranks markets by what a cooperative would actually take home after logistics costs — not just the headline price — helping stakeholders understand where market conditions are genuinely more favourable.

9. Agricultural Financial Risk Engine

A core component of ClimaCrop Intelligence is its ability to translate agricultural conditions into financial risk information, combining climate, crop and market risk factors into a single agricultural risk assessment.



9.1 Composite Agricultural Risk Index

The heart of the financial engine is the Composite Agricultural Risk Index (R_{agri}), a single

quantifiable measure of the risk of financing a farmer or cooperative for a specific crop, county and season:

$$R_{agri} = 0.40 \times \text{Climate Risk} + 0.35 \times (1 - \text{Crop Fitness}) + 0.25 \times \text{Market Volatility}$$

- Climate Risk — derived from rainfall anomaly and dry-spell length; larger deviations increase risk.
- Crop Fitness — the 0–100 suitability score from the crop engine; 100 means perfectly suited.
- Market Volatility — the coefficient of variation of crop prices in the selected market.

The risk score is designed as an agricultural decision-support indicator and should not be interpreted as a conventional credit score.

9.2 Loan Facility Sizing & Risk-Adjusted Pricing

- Eligible Facility = 70% × Production Cost per Acre × Acres — the farmer retains meaningful skin in the game.
- Interest Rate = 12% base rate + a risk premium of 0–8%, scaled by the Composite Risk Index.
- Debt Service Coverage Ratio (DSCR) = Net Farm Profit ÷ Annual Loan Repayment; a DSCR above 1.0 signals comfortable repayment capacity.

9.3 Portfolio Simulation

A portfolio simulator lets banks and SACCOs stress-test their loan book across many borrowers, crops and counties at once, returning:

Financial Indicator	Purpose
Agricultural Risk Score	Measures agricultural exposure
Expected Default Probability	Estimates potential repayment risk
DSCR	Assesses repayment capacity
Facility Sizing	Supports financing decisions
Risk-Adjusted Pricing	Links financing terms to risk
Portfolio Stress Testing	Tests portfolio resilience under climate shocks

10. User Roles & Access Control

The platform implements role-based access control (RBAC) so each user sees only the information relevant to them, built on the same underlying engines.

Role	Demo Username	Default View	Capabilities
Cooperative Member	coop_user	Cooperative Advisory	Crop recommendations, farm profit calculator, market arbitrage, yield projections
Bank / SACCO Officer	bank_officer	Bank & Credit Risk	Loan underwriting, DSCR analysis, portfolio stress testing, risk assessment
Researcher	researcher	Climate Trends	10-year climate trend analysis, rule-based vs. ML benchmarking, read-only Cooperative view
Admin	admin	Cooperative Advisory	All tabs, all persona views, user management

Demo credentials are seeded on first run and persisted to a JSON user store; new accounts can also self-register, making it easy to explore each persona during evaluation.

Security note: authentication currently uses SHA-256 hashing with a single application-wide salt — adequate for a capstone demo, but not for production user data. Future iterations should adopt bcrypt or argon2 with per-user salts.

11. KilimoBot — AI Assistant

KilimoBot is the platform's conversational AI assistant, available to every user role, designed to make the platform's data and analytics accessible to users of all technical backgrounds.

Primary Mode — Google Gemini

- Uses Google's Gemini model via the google-generativeai SDK.
- Seeded with a system prompt describing the platform's exact data scope (116 stations, 35 counties, 40 crops, 5 markets, provenance tags) so answers stay grounded in what the system actually contains.

Offline Fallback Mode

- If no Gemini API key is configured, KilimoBot answers from a built-in rule/keyword knowledge engine.
- Keeps the assistant functional in demonstrations or offline environments, with a reduced conversational range.

The API key resolves in order: a key entered directly in the UI, then `st.secrets["GEMINI_API_KEY"]` for cloud deployments, then a local `GEMINI_API_KEY` environment variable.

12. Technology Stack



Component	Technology	Justification
Language	Python 3.10+	Rich ecosystem for data science, ML and web development
Frontend	Streamlit	Rapid development of data-driven, role-aware dashboards
Data Processing	Pandas	Powerful data manipulation and analysis
Machine Learning	Scikit-learn	K-Means clustering, Random Forest classification
AI / LLM	Google Gemini SDK	Conversational AI for KilimoBot
Authentication	Custom (SHA-256)	Simple, lightweight for demo deployment
Deployment	Docker + Render	Containerized for consistency; easy cloud deployment

13. Deployment & Accessibility

13.1 Live Demo

The platform is deployed and accessible at:

```
https://climacrop-intelligence.onrender.com/
```

13.2 Local Installation

- Prerequisites: Python 3.10+ and Git.
- Clone: `git clone https://github.com/CarsonSila/ClimaCrop-Intelligence.git`
- Create & activate a virtual environment, then: `pip install -r requirements.txt`
- Launch: `streamlit run app.py`, then open `http://localhost:8501`

13.3 Docker Deployment

```
docker build -t climacrop-intelligence .
docker run -p 8501:8501 -e GEMINI_API_KEY="your-key-here" climacrop-intelligence
```

The container reads \$PORT (defaults to 8501), matching Render's deployment convention as well as local Docker usage.

13.4 Test Credentials

Role	Username	Password
Cooperative	coop_user	kilimo2025
Bank / SACCO Officer	bank_officer	sacco2025
Researcher	researcher	tahmo2025
Admin	admin	admin2025

14. Testing & Quality Assurance

14.1 Automated Testing

- `test_auth.py` — demo account login, user registration, invalid-credential handling, role-based access to different views.
- `test_pipeline.py` — verifies all 40 crops in the database, all 200 market-price rows (40 crops × 5 markets), and end-to-end engine output correctness.

```
python -m unittest discover -s tests -p "test_*.py"
```

14.2 Data Provenance as Quality Control

The provenance system doubles as a built-in quality control mechanism, letting users distinguish measured data from modeled predictions, make informed trust decisions, spot where data quality needs improvement, and track data lineage for audit purposes.

15. Key Results

- Climate data can be transformed into regional agricultural insight, not just a rain forecast.
- Crop suitability can be assessed transparently from environmental conditions using auditable rules.
- Market information can be incorporated into agricultural decision-making via transport-adjusted net margins.
- Climate, crop and market factors combine cleanly into one agricultural risk assessment banks can act on.
- The platform provides distinct, purpose-built decision-support views for cooperatives and financial institutions from the same data core.

16. Limitations & Known Issues

Legacy ML Suitability Model

The Random Forest classifier is trained on synthetic samples generated from the same agronomic thresholds it is scored against — a circularity that makes it unsuitable as a primary source of truth. The system defaults to the rule-based engine and treats ML outputs as audit-only.

Static Market Data

Market prices are currently a static snapshot rather than a live feed; refreshing requires running the market-price ingestion module against a real AMIS / KACE / RATIN source.

Demo-Grade Authentication

SHA-256 hashing with a single hardcoded salt is fine for a capstone demo but not for production

user data; bcrypt or argon2 with per-user salts is recommended before real deployment.

UI County List Mismatch

The historical dataset covers 35 counties, but the current Streamlit sidebar offers a curated list of 26; the remaining counties can be added by extending the county list in the app.

Yield Outcome Model Not Yet Wired

A gradient-boosted YieldOutcomeModel exists as scaffolding but is intentionally gated on real outcome data — it will not train on synthetic or estimated data, and is not yet wired into the live app.

17. Future Roadmap

17.1 Data Enhancements

- Live market-price integration from AMIS, KACE or RATIN, moving from snapshot to real-time pricing.
- Full 35-county coverage in the user interface.
- Real yield-data integration from KALRO, census or farmer-reported outcomes to train an independent ML signal.

17.2 Technical Enhancements

- Production-grade authentication (bcrypt / argon2 with per-user salts).
- Deeper data-lineage tracking across transformations, not just at the point of generation.
- A REST API layer so third-party systems can integrate with the advisory and financial engines directly.

17.3 Feature Additions

- A mobile application to reach farmers and cooperative officers with limited internet access.
- Multi-country expansion, beginning with Tanzania, Uganda and Ethiopia.
- Weather-indexed agricultural insurance integrated alongside loan products.
- Carbon-credit estimation for climate-smart farming practices, opening a new revenue

stream for smallholders.

18. Conclusion

ClimaCrop Intelligence represents a significant step forward in climate-smart agricultural decision support for Kenya. By bridging localized climate data, crop suitability analysis and institutional credit underwriting, the platform addresses real needs for farmers, cooperatives and financial institutions alike.

Its commitment to data provenance and transparency sets it apart from conventional advisory tools: users don't just receive a recommendation, they receive the context and confidence needed to act on it — essential in agricultural finance, where trust in data drives institutional adoption.

The modular architecture and role-based access control provide a solid foundation for future growth, and the inclusion of optional live-data connectors and a real-data-only ML pipeline show a platform designed to mature as more data becomes available. As climate volatility continues to challenge Kenyan agriculture, tools like ClimaCrop Intelligence can help put actionable insight directly into the hands of the people who need it most — farmers planting, cooperatives advising, and banks lending.

19. References

- GitHub Repository: github.com/CarsonSila/ClimaCrop-Intelligence
- Live Application: climacrop-intelligence.onrender.com
- Data Sources: TAHMO (Trans-African Hydro-Meteorological Observatory), NASA POWER, FAOSTAT, KNBS (Kenya National Bureau of Statistics), FAO (Food and Agriculture Organization)